## Conditional statements

1 Determine whether or not each of the following is a conjecture.

a Good.

b He is never going to get here.

c When does the next season come out?

d That bridge over there.

e The negative of $x$.

f How do we prove this?

g Dividing by zero is undefined.

h Axes.

i Colorless green ideas furious.

j What is your favourite color?

k Spider silk is stronger than steel.

l Don't sit too close to the screen.

m Welcome to all the new residents.

n Apples are made out of sunshine.

o Curved straight triangular.

p Corresponding angles are congruent.

q A triangle sometimes has five sides.

r Mark the point of intersection.

s Assume this is true for now.

t Who invented geometry?

u What does this button do?

v Yesterday when it all happened.

w The music is loud and the room is crowded.

x The moon orbits the earth.

2 Determine whether or not each of the following is a conditional statement.

a The moon orbits the earth.

b The music is loud and the room is crowded.

c If he replies to me, then we won't be late.

d You can always draw a line parallel to another line.

e The line intersecting two parallel lines.

f The multiplication of two numbers.

g Only equilateral triangles have three sides of equal length.

3 For the following conditional statements, write an equivalent statement.

a Parallel lines never intersect.

b No cat can speak French.

4 Determine whether or not each of the following statements is a conclusion.

a We won't be late.

b He replies to me.

5 Write the negation of the following statements:

a All birds are able to fly.

b Some turtles do not have claws.

c No Libras are Geminis.

d No bicycles have four wheels.

e All houses are connected to the internet.

6 Consider the conditional statement "If $C$ then $D$", where $C$ and $D$ are both conjectures:

a Write the converse of the statement.

b Write the inverse of the statement.

c Write the contrapositive of the statement.

7 Write the converse of the following statements:

a If an animal is a bear, then it has claws.

b If a boomlug is round, then it is delicious.

c If a figure is a square, then it is a quadrilateral.

8 Write the contrapositive of the following statements:

a If an animal is a bird, then it has feathers.

b If a number ends in zero, then it is divisible by ten.

c If a bamboozle is square, then it is loud.

9 Write the inverse of the following statements:

a If an animal is a bird, then it has feathers.

b If two angles form a right angle, then they are complementary.

c If a heffelump is lost in the woods, then it is a woozle.

10 Complete the statement below.
The hypothesis of a conditional is the conclusion of its ▯.

- converse
- inverse
- negation
- contrapositive

## Additional questions

**11** Write a conditional statement using a hypothesis $p$ and conclusion $q$.

**12** Does a conditional statement need to be true?

**13** Identify which of the following are conditional statements.

- **a** If a polygon has three sides, then it is a triangle.
- **b** The church is on Delaney Avenue.
- **c** If I go to school in Florida, then I live in Orlando.
- **d** If it is raining, then we cannot play baseball.
- **e** If a parallelogram is not a rhombus, then it is not a square.
- **f** Red and green are composite colors.

**14** For each of the following conditional statements, identify the hypothesis and conclusion.

- **a** If two angles are vertical, then they are congruent.
- **b** If the sum of a polygon's interior angle measures is $180\degree$, then the polygon is a triangle.
- **c** Two triangles are congruent if they have the same side lengths.
- **d** When two lines are parallel, the consecutive interior angles formed by a transversal are supplementary.

**15** Consider the conditional statement "If I can paint a picture, then I am an artist". Write the:

- **a** Converse statement
- **b** Inverse statement
- **c** Contrapositive statement

**16** For each pair of conditional statements, determine if they are converse, inverse or contrapositive statements.

- **a** If two numbers are even, then their sum will be even.
  If the sum of two numbers is even, then the two numbers are even.
- **b** If two lines do not meet at a right angle, then they are not perpendicular.
  If two lines are perpendicular, then they meet at a right angle.
- **c** If two triangles are similar, they must have three congruent angles.
  If two triangles are not similar, they cannot have three congruent angles.

**17** Determine if each conditional statement is [redacted] false. If it is false, give a counterexample.

- **a** If I need to go shopping, then I will go to Walmart.
- **b** If I have a part-time job, then I am part of the workforce.
- **c** If I score a run in baseball, then I have touched all four bases.
- **d** If I go to school, then I am a student.

**18** Consider the statement "If $x = 2$, then $x^2 = 4$".

- **a** Is this statement true? If not, explain why.
- **b** Is the converse of this statement true? If not, explain why.
- **c** Is the inverse of this statement true? If not, explain why.
- **d** Is the contrapositive of this statement true? If not, explain why.

**19** Write a conditional statement which is true and its converse is not true. Give an example for why the converse is not true.

**20** Consider the statement for the vertical angles theorem:
"Vertical angles are congruent."

- **a** Write the theorem as an "if...then" statement.
- **b** Write a conditional statement for the contrapositive of the vertical angles theorem.

**21** Rewrite the following as conditional statements.

- **a** You miss $100\%$ of the shots you don’t take.
- **b** The bigger they are, the harder they fall.
- **c** When in doubt, tell the truth.